

| Sr.No | Free vibrations                                                                                                                         | Forced vibrations                                                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 2     | To start free vibrations only, the force is required initially.                                                                         | Continuous external periodic force is required. If external periodic force is stopped, then forced vibrations also stop. |
| 3     | The frequency of free vibrations depends on the natural frequency.                                                                      | The frequency of forced vibrations depends on the frequency of the external periodic force.                              |
| 4     | The energy of the body remains constant in the absence of friction, air resistance, etc. Due to damping forces, total energy decreases. | The energy of the body is maintained constant by the external periodic force.                                            |
| 5     | The amplitude of vibrations decreases with time.                                                                                        | Amplitude is small but remains constant as long as an external periodic force acts on it.                                |
| 6     | Vibrations stop sooner or later depending on the damping force.                                                                         | Vibrations stop as soon as external periodic force is stopped.                                                           |

Q 2.4

Q 2.5

Q 2.6

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Students should refer to the answer according to their questions.

Concept: **Free and Forced Vibrations**